

Vector Calculus, Linear Algebra and Differential Forms: A Unified Approach - Solutions

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Chapter 0

Preliminaries

0.2 Quantifiers and Negation

Exercise 0.2.1 Negate the following statements:

- Every prime number such that if you divide it by 4 you have a remainder of 1 is the sum of two squares.
- For all $x \in \mathbb{R}$ and for all $\epsilon > 0$, there exists $\delta > 0$ such that for all $y \in \mathbb{R}$, if $|y - x| < \delta$, then $|y^2 - x^2| < \epsilon$.
- For all $\epsilon > 0$, there exists $\delta > 0$ such that for all $x, y \in \mathbb{R}$, if $|y - x| < \delta$, then $|y^2 - x^2| < \epsilon$.

SOLUTION

- There exists a prime number in the form of $4n + 1$, for $n \in \mathbb{N}_0$ that cannot be expressed as the sum of two squares.
- There exists $x \in \mathbb{R}$ and $\epsilon > 0$ such that for all $\delta > 0$, there exists $y \in \mathbb{R}$ with $|y - x| < \delta$ but $|y^2 - x^2| \geq \epsilon$.
- There exists $\epsilon > 0$ such that for all $\delta > 0$, there exists $x, y \in \mathbb{R}$ with $|y - x| < \delta$ but $|y^2 - x^2| \geq \epsilon$.

Exercise 0.2.2 Explain why one of these statements is true and the other is false:

$$\begin{aligned} &(\forall \text{ man } M)(\exists \text{ woman } W) \mid W \text{ is the mother of } M \\ &(\exists \text{ woman } W)(\forall \text{ man } M) \mid W \text{ is the mother of } M \end{aligned}$$

SOLUTION

The difference between the two statements lies in the quantifier. Obviously the first statement is true while the second is false.

The first statement states that for every man, we can find a woman that is the man's mother. This is inherently true as every man must have a mother.

On the other hand, the second statement states that we can find a woman that is the mother of every man, which is obviously not possible.

0.3 Set Theory

Exercise 0.3.1 Let E be a set, with subsets $A \subset E$ and $B \subset E$, and let $*$ be the operation $A * B = (E - A) \cap (E - B)$. Express the sets

$$\text{a. } A \cup B \quad \text{b. } A \cap B \quad \text{c. } E - A$$

using A , B and $*$.

SOLUTION

Before solving this problem, consider the special case of the operator $*$:

$$X * X = (E - X) \cap (E - X) = X^c \cap X^c = X^c$$

- a. We can first simplify the operator with $A * B = A^c \cap B^c = (A \cup B)^c$. Using the special case, we get

$$(A * B) * (A * B) = (A \cup B)^c * (A \cup B)^c = A \cup B$$

- b. Working backwards, we get

$$A \cap B = (A^c \cup B^c)^c = ((A * A) \cup (B * B))^c = (A * A) * (B * B)$$

- c. This case is the equivalent to the special case above, hence $E - A = A^c = A * A$.